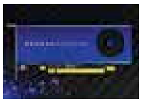

GPU hacked to steal PC data
A GPU's shader clock was turned into a tunable radio transmitter to steal data from an airgapped PC. **https://dgit.in/may20-91**

AI in medical devices
A Q&A on how companies can integrate AI into their medical devices, applications, and the design process. **https://dgit.in/may20-92**

NUTS FOR CORES

Agent001 has a harrowing time explaining CPU cores to a gaming enthusiast

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Once in a blue moon, or should I say once in a process node change, there will be a gamer who wants a ridiculous number of cores for gaming just because the processor manufacturing company advertises it in such a manner. Explaining how they're most probably just going to blow their money on a futile endeavor takes some decent effort. Here is my latest effort at explaining the same to a gamer. If you're sensing a hint of condescension in my tone when I use the term gamer, then you're reading this wrong. I honestly, do not bear *sigh* any ill will even though it might seem that way.

So my friend, let's call him GamerDaddy555, felt that his 3-year old PC has run its course and it's time for him to switch up to the latest generation. Three years seems a decent amount of time for a gaming PC to warrant an upgrade so I was naturally curious about the configuration in the hopes of discovering that all he needs is a graphics card upgrade. Turns out it was a laptop powered by an Intel Core i7 7700HQ. One of the four core mainstream processors when four was all that you could get on mobile devices. While the processor is still quite powerful for a laptop, the desktop processors have steamrolled into some ridiculous core counts. Subsequently, the whole "creator" fad has arisen to showcase how these new machines are capable of multitasking. This is where my friend started having some FOMO and he wanted to shift to a desktop and get a 16-core processor for playing the latest games. I had to step in, for his wallet's sake.

About three years ago, the optimal core count for video games was four, so his Intel Core i7-7700HQ powered laptop was more than sufficient. Since then video game engines have gotten a shot in the arm and now most games make the most of six of your CPU cores. There are certain games that will take up as many cores as you can give it but for the vast majority of games out there and the ones that are scheduled to come out for the next two years, six cores is the magic number. Having more cores is obviously advantageous but if your sole purpose is to play video games for the foreseeable future then an eight-core processor is what you should be looking at, in case game engines become more optimised for using processors in the next two years. Video games are heavily biased towards core frequency rather than core count. After taking up six or eight cores, you're going to begin seeing diminishing returns unless you start increasing the frequency of your cores.


Credit: Ricardo Vallejo Vieira / Shutterstock.com

Notice that I said cores and not processor. That's because the turbo boost frequencies mentioned on the box and packaging materials often